

Working Groups (all)

GY457 Autumn Term 2025 (Seminar Topic 8)

Lecturer: *Dr Sara Bagagli*

Context

This seminar builds on the lecture on Topic 8: “Neighborhoods and segregation.” In the lecture, we discussed the causes and consequences of segregation in urban areas and its effects on neighborhood dynamics and housing markets. The seminar will extend those concepts, focusing on their relevance for real estate. Through discussions and data analysis, students will explore why segregation matters for real estate professionals, measure segregation patterns over time, and examine how these relate to property values.

“Homework” to be carried out prior to the seminar

The compulsory reading for this seminar is:

Cutler, D. M., Glaeser, E. L., & Vigdor, J. L. (1999). The Rise and Decline of the American Ghetto. *Journal of Political Economy*, 107(3): 455-506.

All students must be prepared to answer the questions during the seminar. It is recommended that they prepare some notes. However, no submission of written answers is expected.

Presenters, in addition, are expected to prepare slides for a joint presentation of their answers (one presentation per seminar). In doing so, they may split the tasks or work together on all tasks as they prefer.

Seminar tasks:

- 1) Segregation can influence neighborhood dynamics, affecting property values, access to amenities, and overall demand. For real estate professionals, understanding segregation is not only about market analysis but also about promoting responsible, equitable investment practices that support neighborhood stability and growth. In this introductory part of the seminar, we will discuss why understanding city segregation patterns may be relevant for real estate professionals.

Consider the following as you make your argument:

- a. How might segregation affect property values, housing demand, and investment opportunities?
- b. Why is it essential for real estate professionals to consider the ethical implications of segregation in their work?

- c. How could knowledge of segregation patterns help real estate professionals contribute positively to diverse, inclusive communities?

In your response, think about how real estate decisions can impact neighborhood stability and long-term investment potential. As part of this discussion, consider policies or practices that could help mitigate the negative impacts of segregation on communities. If helpful, you may want to rely on reputable sources (academic research, government reports, newspaper articles...) to support your insights.

- 2) Understanding segregation patterns requires reliable measures that capture the degree of separation or concentration of different racial or ethnic groups across neighborhoods. In this question, we will explore two commonly used indices to measure segregation – the *dissimilarity index* and the *isolation index* – and discuss their relevance for assessing segregation in urban areas.

The *dissimilarity index* measures the evenness with which two groups (e.g., racial or ethnic groups) are distributed across neighborhoods. A higher dissimilarity index suggests that neighborhoods are more segregated.

The *isolation index* measures the likelihood that members of a particular group are exposed to others from the same group within a neighborhood. Unlike the dissimilarity index, the isolation index focuses on group concentration rather than evenness.

- a. Explain the two indices and their formulas, referring to Cutler et al. (1999).
 - b. Why is it important to consider them both? What can be the main advantages of using these indices? Discuss what different insights each index can offer about neighborhood composition.
 - c. Intuitively discuss how each of them might be useful in real estate and what different purposes they might serve.
 - d. Using the dataset GY457_Topic_8_Seminar_Data, calculate the dissimilarity and isolation indices for Chicago between 1920 and 2010. The data is from Bagagli (2025). A description of the dataset and variables is provided at the end of the seminar outline. You may want to briefly explain the code to build those indices to the class. Plot the results over time in two separate graphs and comment on your findings. Does the data for Chicago roughly replicate the trends observed in Figures 1 and 2 in Cutler et al. (1999)?
- 3) Using 100 years of data on the City of Chicago's neighborhoods, we will now explore the relationship between neighborhood property values and demographic composition. Use the dataset GY457_Topic_8_Seminar_Data to answer the following points.
 - a. Let's start with some descriptives. Plot in separate graphs: (i) the share of Black population living in Chicago in each decade over time; (ii) the total population living in the city of Chicago over time and the total Black population living in Chicago over time (you may want to plot these two together); in addition: the city-level average measures of (iii) house value, (iv) rent, (v) land value, and (vi) the share of college graduates over time.

Census data did not collect all information for all variables across all periods, so that some trends will be shorter than others. Comment on these city trends.

- b. Pooling together all the data available across all years, plot the relationship between neighborhood value (using one of the measures available) and demographic composition. I suggest using the *binscatter* command, to make the relationship clearer.¹ What do you find? What can be the reason(s) behind this non-linear relationship? Discuss.
- c. Investigate again the same relationship as in point 3(b), but separately by decade. In particular, given the small share of Black homeowners – especially in the earliest periods – you may want to use median rent as a proxy for neighborhood property values. Do you observe any remarkable differences in the relationship between the share of the Black population in the neighborhood and the median rent of its residents over time? If so, pick the three most exemplary decades in your view (eventually, with reference to the three periods of segregation as evidenced in Cutler et al., 1999).
- d. Let's now run a set of regressions to examine the relationship between neighborhood property values and racial composition. To do so, we will explore step by step the use of fixed effects in regression analyses. Note that the analysis does not imply any inherent differences in value due to racial composition but instead aims to examine how complex socioeconomic factors influence housing markets over time.
 - i. Run an OLS regression of house values on the racial composition of neighborhoods (share Black), pooling all data together. How do we interpret this magnitude?
 - ii. Run the same regression but add year fixed effects (*i.year* in Stata) to control for the time trends we observed before. How does the magnitude of the estimated coefficient change?
 - iii. Run the regression in point d(ii) again, adding “city sides” fixed effects. This corresponds to a set of dummy variables (*id_sides*) that captures whether the neighborhood is located in the north, west, or south side of Chicago. Why might it be important to include these controls?
 - iv. Run the regression in point d(iii), adding an interaction term between year and city-side fixed effects.² The interaction terms capture time and city-side-specific trends in house values. This means that the regression now effectively controls how each side of the city has changed over time (for instance, capturing infrastructure developments or neighborhood revitalization efforts

¹ If you do, it might be helpful to also show the results using a normal scatter plot. This might help visualize the advantages of binning observations together to reduce noise, when we have large datasets at hand.

² Interaction between variables in Stata can be generated directly by adding an *ashtag* between the two variables (so in this case, you can simply add *i.year#i.id_sides* to the set of regressors in the regression for d(iii)).

- that differ across these broad locations). How does the magnitude of the estimated coefficient change? What could be the reason(s)?
- v. Let's now add some neighborhood-level controls. You can choose which one(s) you find more appropriate among the ones available in the dataset. Motivate your choice and discuss your findings.
 - vi. Finally, run the same regression as in point d(v) but adding a high-dimensional number of neighborhood-specific fixed effects. Using fixed effects is a great tool to account for time-invariant differences between neighborhoods that could influence house values but are not directly observable. For instance, certain neighborhoods can have historical reputations and unique characteristics that do not change over time. Neighborhood fixed effects control for these constant characteristics by "differencing out" each neighborhood's unique attributes. Given the large number of fixed effects (one for each neighborhood), install the Stata command *reghdfe*.³ What do you find? Why have some of the variables now dropped out of the regression?
 - vii. Discuss the key insights of your analysis. Despite using fixed effects, the regression is far from having a causal interpretation. What are some of the remaining threats to identification? Why should real estate professionals interpret these results with caution, avoiding oversimplified conclusions?

During the seminar

After an introduction and discussion of pressing questions, the presenters will walk the group through their answers during an "interactive" presentation. All presenters should carry an approximately similar weight in the presentation. The presentation, in total, may cover up to 60 minutes (e.g., 15 min for 1), 20 min for 2), 20-25 min for 3)). During the presentation, the presenters are expected to "engage" with the audience, i.e., prompt the audience to suggest answers. Likewise, the audience is expected to challenge the answers provided by the presenters. The lecturer will monitor the presentation and the discussion and intervene if appropriate. The presenters are expected to make notes during the discussion in case corrections to their suggested answers are due.

After the seminar

If indicated by the feedback during the seminar, presenters will revise their presentation. Within 24 hours, they will ask the instructor to upload the slides onto the GY457 Moodle page. Please use the following filenames to save your presentations: Topic_8_Group_X, where X indicates the seminar group.

³ The command requires the option *absorb()* to function which will tell Stata to control (absorb) neighborhood fixed effects (in this case, the implementation should require *absorb(geo2010)*).

VARIABLES DESCRIPTION (All variables are at the neighborhood level)

The dataset contains information for each neighborhood (census tract) in Chicago from decadal censuses between 1920 and 2010. Included are the following variables. Please note that some values may be missing for certain neighborhoods and years, either because the Census did not collect information for all variables in each period or because some neighborhoods had not yet been established in the earlier periods under analysis.

black	Number of Black residents
tract_sblack	Share of Black residents
pop	Residential population
popd	Population density
tr_hu_avgvalr2	Average real house value (\$2010)
mrentr2	Median rent, real (\$2010)
lvr2	Land value, real (\$2010)
llvr2	Log land value, real (\$2010)
col	Number of college graduates
pcol	Share college graduates (out of people 25 and above)
d2cbd	Distance from CBD (meters)
ld2cbd	Log distance from CBD
d2shore	Distance to closest shoreline (meters)
ld2shore	Log distance to closest shoreline
d2highway	Distance from closest current highway (meters)
ld2highway	Log distance to closest current highway
d2rail	Distance to closest historical rail (meters)
ld2rail	Log distance to closest historical rail
tr_hu_avgage	Average age of housing units
tr_hu	Number of housing units
id_sides	Categorical variable for being in north, west, south
geo2010	Neighborhood ID
year	year